3	W103	Jammed	Alex

Exercise 6
Advanced SQL
DATE functions

1. Add 6 months to each employees hire date using DATEADD()

```
Select emp_id, name, hire_date,
DATEADD(month, 6, hire_date) As hire_plus_6_months
From Employees;
```

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

② Use DATEDIFF() to find age in days from dob to today

```
Select studentid, name,
    DATEDIFF(Day, 2026-04-15, dob) As age_in_days
from Students;
```

Student_id	name	age_in_days
101	Maya	7510
102	Ethan	7808
103	Sienna	7335

3 Find how many days are left until each until each event using DATEDIFF()

```
Select event_id, event_name,
DATEDIFF(Day, 2026-04-15, event_date) AS days_remaining
From Events ;
```

event_id	event_name	days_remaining
2	Workshop	227
3	Hackathon	450

4 Calculate the number of days between issue_date and due_date.

```
Select invoice_id, issue_date, due_date,
Datediff (day, issue_date, due_date) As days_between
From Invoices ;
```

Invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

5 Format start_date as 'Month YYYY' using To-Char()

```
Select course_id, name,
To_Char(start_date, 'FMMonth YYYY') As formatted_date
from Courses ;
```

Course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

6 Create full date from parts using DATE_FROM_PARTS()

```
Select member_id
DATE_FROM_PARTS(start_year, start_month, stat_day) AS
        full_start_date
From Memberships ;
```

member_id	full_start_date
1	2023-05-10
2	2022-11-25

7 Extend each renewal_date by 1 year using DATEADD()

```
Seled Sub_id, plan,
DATEADD(YEAR, 1, renewal_date) As extended_renewal_date
From Subscriptions;
```

Sub_id	plan	extended_renewal_date
11	Basic	2026
12	Premium	2026

8 Show current date and difference from order_date. Use current_Date and Datediff()

```
Select Current date;
```
2026-04-15

```
Seled order_id, order_date,
DATEDIFF(day, 2026-04-15, order_date) As days_since_order
from Orders;
```

Order_id	order_date	today_date	days_Since_order
1001	2025-04-15	2026-04-15	365
1002	2025-04-10	2026-04-15	370

9 Trainings:

Q9. Extract the year from training_date using Date_Part or Extract.

```
Select training_id, topic,
DATE_PART(Year, training_date) As training_year
From Trainings;
```

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

10 Extract hour and minute from published_on

Select post_id, title,
Extract (hour from published_on) As hour_published,
Extract (minute_publ

Select post_id, title,
Extract (hour from published_on) As hour_published,
Extract (minute from published_on) As minute_published
From Blog_Posts ;

post_id	title	hour_published	minute_published
1	SQLTips	10	15
2	Data Cleaning	16	45

11 Calculate days left until license expiry using DATEDIFF and todays date

Select Current_Date ;
2026-04-15
Select driver_id, license_expiry,
DATEDIFF(day, 2026-04-15, license_expery) As days_left
from Drivers ;

~~Driver_id~~

driver_id	license_expiry	days_left
301	2025-08-10	248
302	2023-12-31	835

12 Display the current timestamp and calculate seconds since the message was sent.

Select message_id, sent_timestamp

Select Current_timestamp;

current_timestamp
2026-04-15 15:02:55

Select message_id, sent_timestamp,
DATEDIFF(Second, Current_timestamp, Sent_timestamp) AS
seconds_since_sent
From Messages ;

message_id	Sent_timestamp	Current_timestamp	Seconds_since_sent
1	2025-04-19 09:32:45	2026-04-15 15:02:55	31468210
2	2025-04-18 23:59:59	2026-04-15 15:02:55	31502576

13 Add 15 days to return_date using DATEADD() to show restock_date

Select return_id, return_date,
DATEADD(day, 15, return_date) As restock_date
from Returns ;

return_id	return_date
901	2025_04_20
902	2025_04_16

14 Convert assigned_on to date using TO_DATE() (if its stored as string)

```
Select assign_id,
    To_Date(assigned_on) As assigned_on_date
From assignment;
```

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

15 Convert scheduled_time to formatted string like 'April 19, 2025 at 2:00 PM' using TO_CHAR()

```
Select meeting_id,
    TO_CHAR(scheduled_time, Month DD, YYYY at HH:MI AM') AS
        formatted_meeting_time
From meetings;
```

meeting_id	formatted_meeting_time
1	April 19, 2025 at 02:PM
2	April 19, 2025 at 09:30 AM